



Technical Service Bulletin: TSB160020

Released Date: 07-Mar-2016

New Cylinder Head Capscrews and Torque Specification Revision

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document combines and summarizes the following Service Parts Topics and Technical Service Bulletins:

- 89T2-4 (1989)
- 00T2-5 (2000)
- 05t2-7 (2005), Converted to TSB101651
- 05t2-10 (2005) Converted to TSB101660
- 05t2-8 (2005) Converted to TSB101652
- 05t2-9 (2005) Converted to TSB101653

Product Affected

- ISM
- ISM CM876 SN
- L10
- L10 Gas
- M11
- QSM11

Description of Change

This document introduces new cylinder head capscrews. The new capscrews are a flange head style capscrew and eliminate the need for the hardened washer, Part Number 3019631.

The installation procedure for cylinder head capscrews has also been revised to the torque plus angle method.

Reason for Change

The new capscrews have improved corrosion resistance and the torque plus angle installation procedure produces higher and more consistent clamp load for the cylinder head gasket joint.

Service Instructions

The following torque plus angle installation procedure applies **only** to flange head style capscrews without a hardened washer.

All 32 capscrews **must** be rotated 180 degrees after having been torqued to 136 N•m [100 ft-lb]. Make a mark on the cylinder head as a reference. Rotate the capscrew until the mark on the cylinder head is more than two flats but less than three flats.

When using torque plus angle, the tolerance on the 180 degree angle of rotation is between two and three flats. If the capscrew is unintentionally rotated beyond three flats, do **not** loosen the capscrew. The clamp load is still acceptable. However, rotating the capscrew beyond three flats reduces the number of potential reuses.

Tighten the capscrews in the sequence shown in Figure 4 to the specified values:

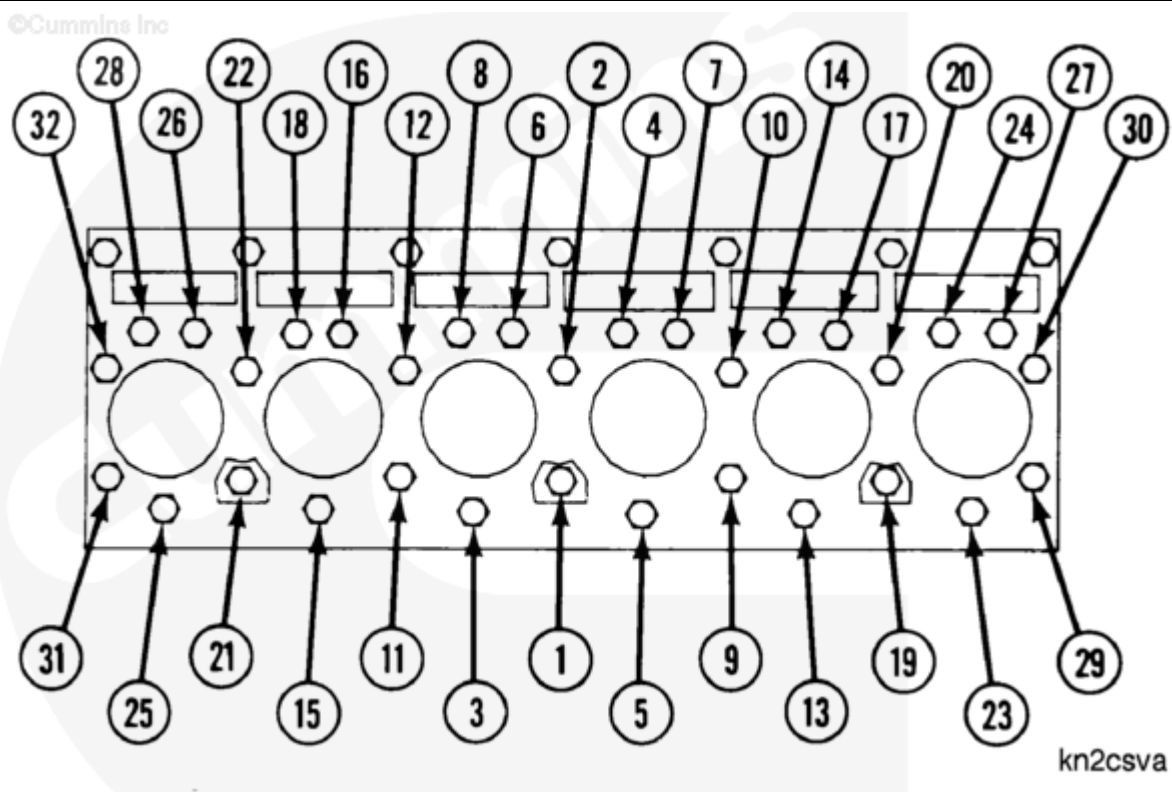


Figure 4, Capscrew Torque Sequence

Torque Value:

1. 68 N•m [50 ft-lb]
2. 136 N•m [100 ft-lb]
3. Rotate 180 Degrees

Note : Rotate at least two flats, but not more than three.

Service Parts Availability

Service parts are available. See Table 1.

Table 1, Service Parts					
Part Description	Existing Part Number	Obsoleted	Superseded	New Part Number	Quantity Per Engine
Short Cylinder Head Capscrew	3019641	Yes	Yes	4022907	3
Long Cylinder Head Capscrew	3019640	Yes	Yes	4923187	29
	3045850				

Part Compatibility

- The hardened washer, Part Number 3019631, must **not** be used with the new flange head capscrews, Part Number 4022907 and 4923187.
- New flange head capscrews and capscrews with hardened washers can **not** be intermixed on the same engine.
- If replacement of a capscrew with a hardened washer is required and a like part number is **not** available, all 32 cylinder head capscrews **must** be replaced with the new flange head capscrews.

Part Identification

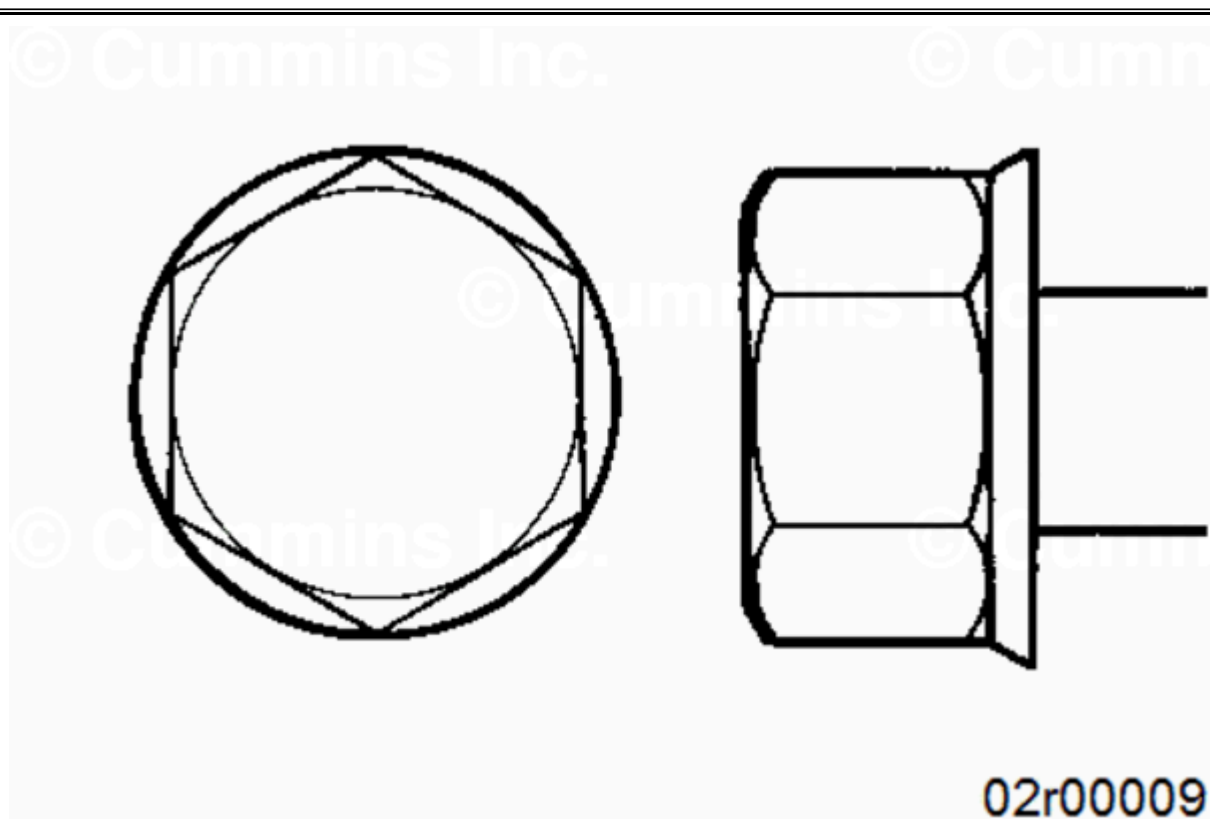


Figure 1, New Flange Head Capscrew.

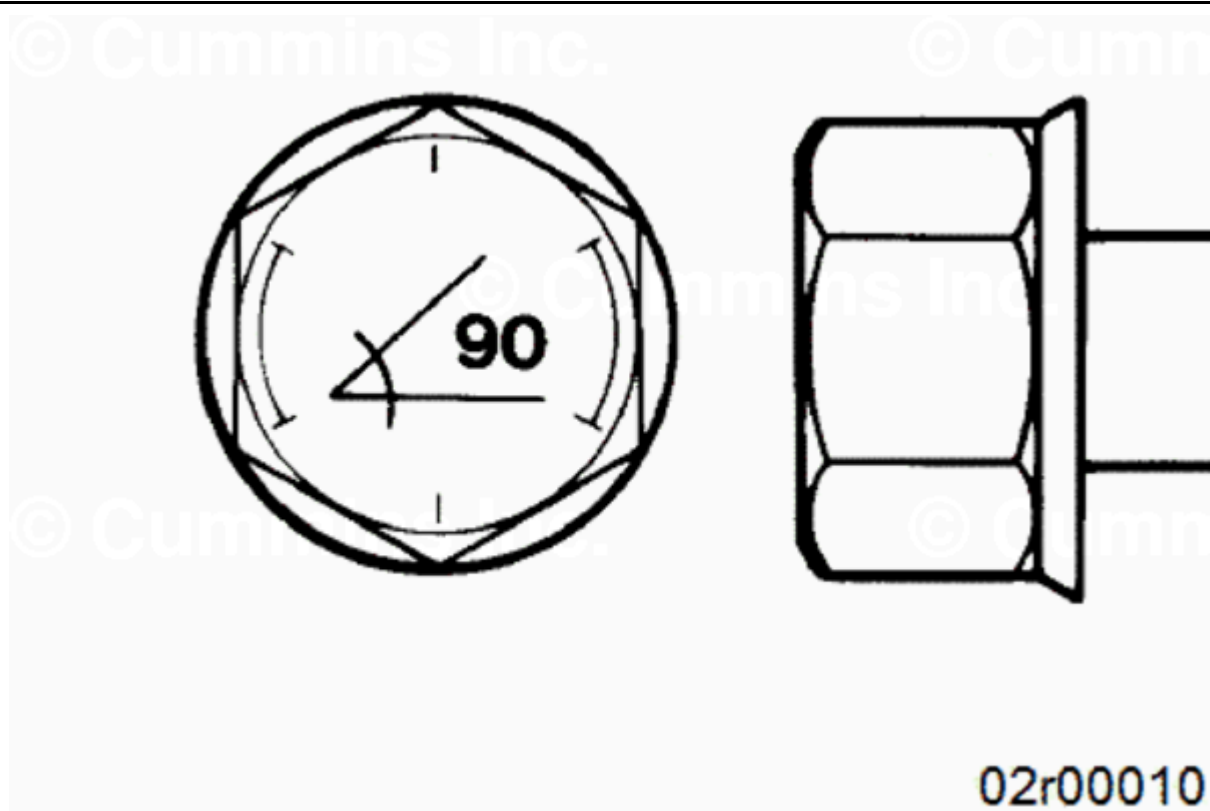


Figure 2, Obsolete Flange Head Capscrew

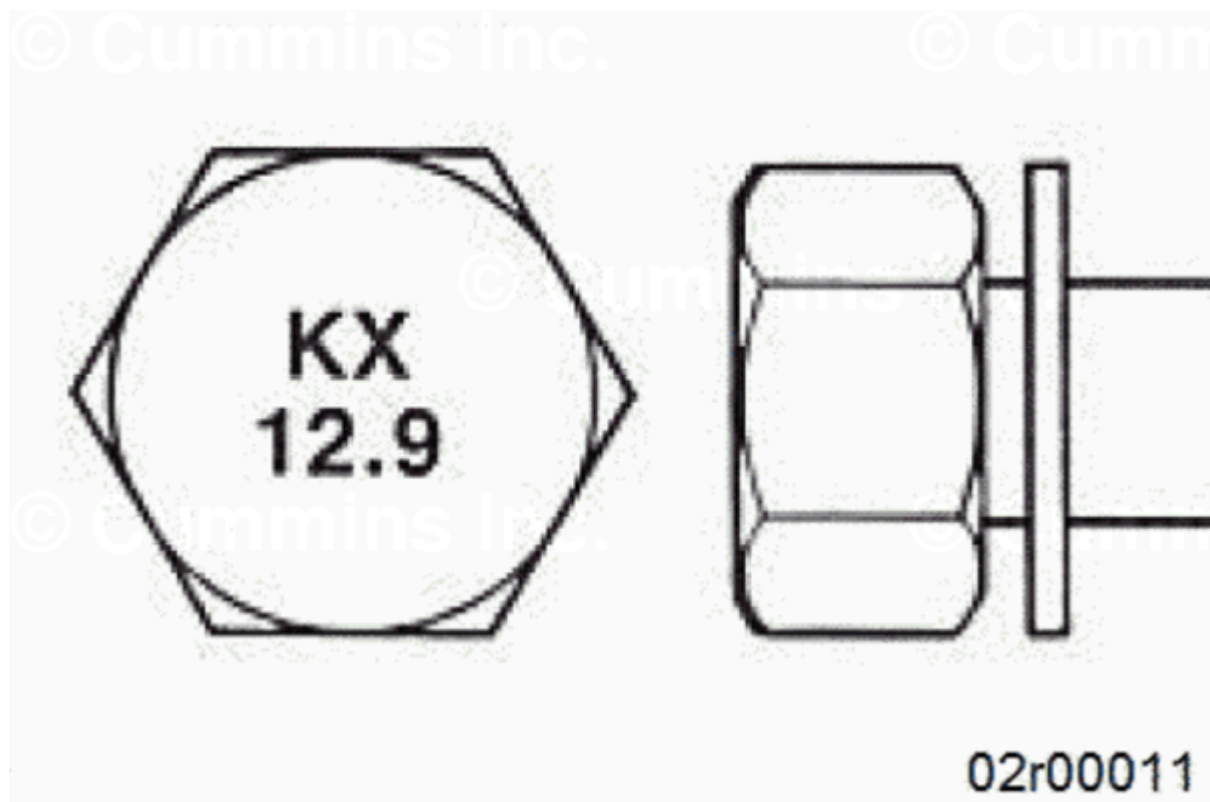


Figure 3, Obsolete Capscrew with Hardened Washer.

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Troubleshooting and Repair	L10	3810439*	Cylinder Head	002-004	2
Troubleshooting and Repair	L10 Gas	3666207	Cylinder Head	002-004	2
Troubleshooting and Repair	L10 (Pre-1994)	3810426*	Cylinder Head	002-004	2
Service Manual	ISM	3666322	Cylinder Head	002-004	2
Service Manual	ISM CM876 SN	4022196	Cylinder Head	002-004	2
Troubleshooting and Repair	M11	3666139	Cylinder Head	002-004	2
*Manual available in print only					

Document History

Date	Details
2016-2-29	Module Created